

Riemannian Geometry and Geometric Field Equations of Two-Agent Interaction Dynamics: From Time-Dependent to State-Dependent Manifolds

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Abstract

This work presents a unified geometric framework for two-agent interaction dynamics, developed in two successive stages. In the first stage, we construct a time-dependent Riemannian metric on the interaction manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$ by imposing a linear constraint $I_P = \gamma(t)I_S$ on the passive interaction variable. The induced metric $G(t) = I + \gamma^T \gamma$ gives rise to a natural interaction flow tensor

$$K^i_j = \frac{1}{2} G^{ik} \dot{G}_{kj},$$

whose time evolution governs all Ricci curvature components. In the second stage, we extend this model to a state-dependent framework by allowing $\gamma = \gamma(t, I_S)$, which introduces spatial variation in the interaction metric

$$G(t, I_S) = I + J^T J, \quad J = \gamma + (\nabla_{I_S} \gamma) I_S.$$

This extension generates non-trivial spatial curvature and leads naturally to a Ricci tensor decomposition into temporal and spatial contributions. Building on this geometric foundation, we introduce an interaction action functional and prove, under the axioms of locality, diffeomorphism covariance, and second-order field equations, that the admissible action is uniquely determined by the scalar curvature. Variation yields the interaction field equation

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu},$$

which is structurally analogous to the Einstein field equations. The framework provides a rigorous nonlinear description of interaction dynamics, where $T_{\mu\nu}$ encodes reinforcement and perceptual structures, offering a unified perspective linking differential geometry with behavioral feedback systems.

Contents

1	Introduction	3
2	Stage I: Time-Dependent Interaction Metric	4
2.1	Interaction Variables and Constraint	4
2.2	Extended Euclidean Construction	4
2.3	Induced Interaction Metric	4
3	Stage I: Curvature from Temporal Deformation	5
4	Stage II: State-Dependent Interaction Metric	6
4.1	Motivation for State Dependence	6
4.2	Extended Interaction Constraint	6
4.3	State-Dependent Metric	6
5	Stage II: Connection and Curvature with State Dependence	7
5.1	Levi-Civita Connection	7
5.2	Ricci Tensor Decomposition	7
5.3	Non-Triviality of the Interaction Geometry	7
6	Interaction Action Functional and Field Equations	8
6.1	Axioms of the Interaction Action	8
6.2	Derivation of the Unique Action	8
6.3	Main Theorem: Uniqueness of the Interaction Action	9
7	Physical Interpretation	10
7.1	Curvature as Interaction Geometry	10
7.2	Role of State Dependence	10
8	Conclusion	11

1 Introduction

Mathematical descriptions of interacting systems often rely on dynamical equations defined in Euclidean state spaces. However, when interaction variables mutually constrain one another, the underlying state space may naturally acquire a nontrivial geometric structure. In such cases, curvature provides a coordinate-invariant measure of structural deformation induced by temporal evolution.

This work develops the geometric theory of two-agent interaction dynamics in two successive stages, motivated by increasing levels of physical realism.

In the *first stage* (Sections 2–3), we consider a time-only interaction rule $\gamma = \gamma(t)$, where a passive interaction variable I_P is linearly constrained by an active variable I_S through a time-dependent matrix transformation. This yields a block-diagonal Riemannian metric whose spatial components evolve purely in time. All curvature then arises from temporal deformation, and the system is equivalent to a cosmological-type metric evolution.

In the *second stage* (Sections 4–6), we generalise to $\gamma = \gamma(t, I_S)$, so that the interaction rule depends on the current interaction state itself. This extension is motivated by many real-world phenomena:

- nonlinear feedback in dynamical systems,
- emotional escalation and saturation in social interactions,
- biological response saturation,
- adaptive feedback control.

The state dependence introduces spatial variation in the metric, generating non-trivial intrinsic curvature even in the absence of temporal deformation. We then formulate a variational principle and derive Einstein-type interaction field equations from first principles.

Throughout, the key quantities are:

- the *interaction flow tensor* K^i_j , governing temporal metric evolution,
- the *emotional quantity tensor* E_{ij} , measuring intrinsic interaction acceleration,
- the *interaction source tensor* $T_{\mu\nu}$, encoding reinforcement and perceptual forcing.

2 Stage I: Time-Dependent Interaction Metric

2.1 Interaction Variables and Constraint

We consider two interaction vectors

$$I_S(t) \in \mathbb{R}^2, \quad I_P(t) \in \mathbb{R}^2,$$

representing active and passive interaction components, respectively. We impose the *linear interaction constraint*

$$I_P(t) = \gamma(t) I_S(t),$$

where $\gamma(t) \in \mathbb{R}^{2 \times 2}$ is a differentiable matrix-valued function of time.

2.2 Extended Euclidean Construction

Consider the extended coordinate space

$$(t, I_P^1, I_P^2, I_S^1, I_S^2) \in \mathbb{R} \times \mathbb{R}^2 \times \mathbb{R}^2,$$

equipped with the standard Euclidean line element

$$ds^2 = dt^2 + dI_P^T dI_P + dI_S^T dI_S.$$

2.3 Induced Interaction Metric

Imposing the constraint gives $dI_P = \gamma(t) dI_S$ at fixed t . Substituting into the Euclidean line element:

$$ds^2 = dt^2 + dI_S^T (I + \gamma(t)^T \gamma(t)) dI_S.$$

We therefore define the *spatial interaction metric*

$$G(t) = I + \gamma(t)^T \gamma(t).$$

By construction $G(t)$ is symmetric, positive definite, and depends only on time. The full metric on the manifold $\mathcal{M} = \mathbb{R}_t \times \mathbb{R}_{I_S}^2$ is

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t) \end{pmatrix},$$

where $a > 0$ is a constant temporal scale factor.

3 Stage I: Curvature from Temporal Deformation

Definition 3.1 (Interaction Flow Tensor). *The interaction flow tensor is*

$$K^i{}_j = \frac{1}{2} G^{ik} \dot{G}_{kj},$$

where $\dot{G}_{kj} = dG_{kj}/dt$. *This tensor measures the rate of temporal deformation of the interaction metric.*

Theorem 3.1 (Non-vanishing Christoffel Symbols). *For the metric above, the only non-vanishing Christoffel symbols are*

$$\Gamma^0_{ij} = -\frac{1}{2a^2} \dot{G}_{ij}, \quad \Gamma^i_{0j} = \Gamma^i_{j0} = K^i{}_j.$$

Proof. From $\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu})$ and the block-diagonal structure with $g_{0i} = 0$, $g_{00} = a^2$ constant, and $\partial_k G_{ij} = 0$, the only surviving terms are those involving $\partial_t G_{ij}$. $\square$

Theorem 3.2 (Ricci Tensor: Temporal Component).

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

Proof. Since $\Gamma^\lambda_{00} = 0$, the first and third terms in the Ricci formula vanish. The second term gives $-\partial_t \Gamma^i_{0i} = -\text{Tr}(\dot{K})$, and the quartic contraction yields $-\text{Tr}(K^2)$. $\square$

Definition 3.2 (Emotional Quantity Tensor). *The emotional quantity tensor is*

$$E_{ij} = \dot{K}_{ij} + K_{ik} K^k{}_j.$$

It describes the intrinsic interaction acceleration independent of collective coupling.

Theorem 3.3 (Ricci Tensor: Spatial Components).

$$R_{ij} = E_{ij} + K_{ij} \text{Tr}(K) = \dot{K}_{ij} + K_{ik} K^k{}_j + K_{ij} \text{Tr}(K).$$

Proof. Since $\partial_k G_{ij} = 0$, all spatial Christoffel symbols vanish. Computing each term in the Ricci formula using only Γ^0_{ij} and Γ^i_{0j} yields the three surviving contributions as stated. $\square$

These results show that in the purely time-dependent model, *all curvature arises from temporal deformation of the interaction metric*. The geometry is structurally analogous to a cosmological (FLRW-type) metric.

4 Stage II: State-Dependent Interaction Metric

4.1 Motivation for State Dependence

If $\gamma = \gamma(t)$ only, then $\partial_k G_{ij} = 0$ and the spatial geometry is flat. Many realistic interaction systems require the interaction rule to depend on the current interaction level:

- *Emotional escalation*: the coupling strength changes as the emotional intensity grows.
- *Biological saturation*: response functions saturate at high stimulation.
- *Adaptive feedback*: the system adjusts its own sensitivity based on state.

We therefore extend $\gamma = \gamma(t)$ to $\gamma = \gamma(t, I_S)$.

4.2 Extended Interaction Constraint

With $\gamma = \gamma(t, I_S)$, the differential of the constraint $I_P = \gamma I_S$ is

$$dI_P = \gamma dI_S + (\partial_t \gamma) I_S dt + (\partial_{I_S} \gamma) (dI_S) I_S.$$

At fixed time ($dt = 0$) we define the *interaction Jacobian*

$$J(t, I_S) = \gamma(t, I_S) + (\nabla_{I_S} \gamma(t, I_S)) I_S, \quad \text{where} \quad (\nabla_{I_S} \gamma)_{ij} I_S^k = \frac{\partial \gamma_{ij}}{\partial I_S^k} I_S^k.$$

4.3 State-Dependent Metric

Substituting into the Euclidean line element gives the induced metric

$$G(t, I_S) = I + J(t, I_S)^T J(t, I_S).$$

This metric is symmetric, positive definite, and *depends on both t and I_S* . The full manifold metric remains block-diagonal:

$$g_{\mu\nu} = \begin{pmatrix} a^2 & 0 \\ 0 & G_{ij}(t, I_S) \end{pmatrix}.$$

5 Stage II: Connection and Curvature with State Dependence

5.1 Levi-Civita Connection

Theorem 5.1 (Connection Structure). *For the state-dependent metric, the non-vanishing Christoffel symbols are*

$$\begin{aligned}\Gamma_{ij}^0 &= -\frac{1}{2a^2}\partial_t G_{ij}, \\ \Gamma_{0j}^i &= K^i_j = \frac{1}{2}G^{ik}\partial_t G_{kj}, \\ \Gamma_{jk}^i &= \frac{1}{2}G^{il}(\partial_j G_{kl} + \partial_k G_{jl} - \partial_l G_{jk}).\end{aligned}$$

The spatial symbols Γ_{jk}^i are non-zero whenever $\nabla_{I_S}\gamma \neq 0$.

5.2 Ricci Tensor Decomposition

Theorem 5.2 (Ricci Tensor Decomposition). *The Ricci tensor decomposes as*

$$R_{ij} = \dot{K}_{ij} + K_{ik}K^k_j + K_{ij}\text{Tr}(K) + R_{ij}^{\text{space}},$$

where R_{ij}^{space} is the Ricci tensor of the spatial metric G_{ij} computed with respect to its own Levi-Civita connection.

Remark 5.1. *The decomposition cleanly separates two sources of curvature:*

1. Temporal curvature: the $\dot{K}$ and K^2 terms arising from evolution of the interaction rule over time.
2. Spatial curvature: R_{ij}^{space} arising from the state-dependent variation $\partial_k G_{ij} \neq 0$.

This geometric splitting is central to the interpretation of emotional escalation and non-linear feedback in the interaction manifold.

Theorem 5.3 (Temporal Component).

$$R_{00} = -\text{Tr}(\dot{K}) - \text{Tr}(K^2).$$

5.3 Non-Triviality of the Interaction Geometry

Theorem 5.4 (Non-trivial Curvature Condition). *If $\nabla_{I_S}\gamma \neq 0$ in some region of the interaction space, then the spatial curvature R_{ij}^{space} is generically non-zero.*

Proof. We have $\partial_k G_{ij} = (\partial_k J)^T J + J^T (\partial_k J)$. If $\nabla_{I_S} \gamma \neq 0$ then $\partial_k J \neq 0$, so $\partial_k G_{ij} \neq 0$. Non-vanishing spatial derivatives of the metric produce non-zero spatial Christoffel symbols, which in turn generate non-zero R_{ij}^{space} . $\square$

The purely time-dependent model ($\gamma = \gamma(t)$) is recovered as the special case $\nabla_{I_S} \gamma = 0$, for which $R_{ij}^{\text{space}} = 0$.

6 Interaction Action Functional and Field Equations

6.1 Axioms of the Interaction Action

To derive a governing equation for the interaction geometry, we impose four physical axioms.

Axiom 6.1 (Locality). *The interaction dynamics at each point depend only on the metric and its derivatives up to finite order at that point:*

$$S[g] = \int_{\mathcal{M}} \mathcal{L}(g, \partial g, \partial^2 g, \dots) d\mu_g.$$

Axiom 6.2 (Diffeomorphism Covariance). *The action is invariant under smooth coordinate transformations $x^\mu \mapsto x'^\mu(x)$:*

$$S[g] = S[\varphi^* g] \quad \forall \varphi \in \text{Diff}(\mathcal{M}).$$

Axiom 6.3 (Metric Dynamics). *The metric $g_{\mu\nu}$ is the sole dynamical geometric field; the Lagrangian depends only on tensors generated from g .*

Axiom 6.4 (Second-Order Field Equations). *The Euler–Lagrange equations derived from the action contain at most second derivatives of the metric, ensuring well-posed interaction evolution.*

6.2 Derivation of the Unique Action

Lemma 6.1 (Covariant Scalar Construction). *Any diffeomorphism-invariant local Lagrangian must be constructed from contractions of the Riemann tensor $R^\rho_{\sigma\mu\nu}$ and its covariant derivatives.*

Lemma 6.2 (Second-Order Constraint). *If the Lagrangian contains ∇R or higher, the field equations contain derivatives of order ≥ 3 in the metric, violating Axiom 4.*

Proof. Since $R \sim \partial^2 g$, any $\nabla R \sim \partial^3 g$, so $\delta S \sim \partial^4 g$, giving fourth-order equations. $\square$

Corollary 6.3. *The admissible Lagrangian depends only algebraically on the curvature: $\mathcal{L} = \mathcal{L}(g, R_{\mu\nu\rho\sigma})$.*

Lemma 6.4 (3D Curvature Identity). *In three spacetime dimensions, the Riemann tensor is completely determined by the Ricci tensor:*

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho} - \frac{R}{2}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}).$$

Every scalar invariant therefore reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.

Lemma 6.5. *Quadratic curvature invariants such as $\int R_{\mu\nu}R^{\mu\nu}\sqrt{|g|}$ generate fourth-order field equations and are therefore excluded by Axiom 4.*

6.3 Main Theorem: Uniqueness of the Interaction Action

Theorem 6.6 (Uniqueness of the Interaction Action Functional). *Let $(\mathcal{M}, g)$ be a three-dimensional interaction manifold with coordinates (t, I_S^1, I_S^2) equipped with the block-diagonal metric induced by the state-dependent interaction rule. Under Axioms 1–4, the unique admissible action functional is, up to a boundary term,*

$$S[g] = \int_{\mathcal{M}} \left(\frac{1}{2\kappa} R + \Lambda \right) \sqrt{|g|} d^3x,$$

where R is the Ricci scalar, κ is a coupling constant, and Λ is a cosmological-type constant. Variation of this action yields the interaction field equation

$$\boxed{R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}},$$

where the interaction source tensor $T_{\mu\nu}$ arises from matter or interaction contributions to be specified in applications.

Proof. The argument proceeds in six steps:

1. **Locality and Covariance.** By Axiom 1–2 and the non-tensorial nature of Christoffel symbols, the Lagrangian must be built from the Riemann tensor and its covariant derivatives.
2. **Dimensional Reduction.** In three dimensions the Weyl tensor vanishes and every scalar invariant reduces to a function of R and $R_{\mu\nu}R^{\mu\nu}$.
3. **Second-Order Constraint.** By Axioms 4 and the preceding lemmas, all terms involving ∇R or quadratic curvature are excluded.
4. **Admissible Lagrangians.** The only surviving scalars are 1 (the volume element) and R (linear in curvature), giving $\mathcal{L} = \alpha + \beta R$.

5. **Normalization.** Setting $\beta = 1/(2\kappa)$ and $\alpha = \Lambda$ produces the stated action.
6. **Variation.** Standard variation of the Einstein–Hilbert term yields the Einstein tensor $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$; the cosmological term contributes $\Lambda g_{\mu\nu}$; and any interaction sources produce $\kappa T_{\mu\nu}$.
7. **Uniqueness.** Any other diffeomorphism-invariant local scalar leading to second-order equations must reduce to the above form in three dimensions.

□

The interaction field equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}$$

is structurally identical to the Einstein field equations of general relativity, but here it governs the geometry of the interaction space rather than spacetime.

7 Physical Interpretation

7.1 Curvature as Interaction Geometry

The two-stage framework provides a geometric dictionary:

Geometric object	Interaction interpretation
Metric $G_{ij}(t, I_S)$	Structure of the interaction space
K^i_j	Rate of change of interaction coupling
E_{ij}	Intrinsic interaction acceleration
R_{00}	Negative trace of total interaction acceleration
R_{ij}	Spatial interaction curvature
$T_{\mu\nu}$	Reinforcement and perceptual sources
Λ	Baseline interaction intensity

7.2 Role of State Dependence

State dependence $\nabla_{I_S}\gamma \neq 0$ is essential for three reasons:

1. **Physical realism.** Real interaction systems exhibit state-dependent coupling (emotional escalation, biological saturation, adaptive feedback).

2. **Mathematical completeness.** Without state dependence the spatial geometry is flat ($R_{ij}^{\text{space}} = 0$) and the model reduces to Stage I.
3. **Field equation structure.** Non-trivial spatial curvature is necessary for the interaction field equation to have meaningful non-flat solutions.

8 Conclusion

This paper has developed a unified geometric formulation of two-agent interaction dynamics in two stages, culminating in an Einstein-type interaction field equation derived from first principles.

In **Stage I**, we constructed a time-dependent Riemannian metric from a linear interaction constraint $I_P = \gamma(t)I_S$. The interaction flow tensor K^i_j and the emotional quantity tensor E_{ij} emerge naturally as geometric quantities governing Ricci curvature. All curvature in this stage arises from temporal deformation, analogous to cosmological metric evolution.

In **Stage II**, we generalised to $\gamma = \gamma(t, I_S)$, introducing spatial variation in the metric and non-trivial intrinsic curvature. The Ricci tensor decomposes cleanly into temporal and spatial contributions, with the spatial term generated by state dependence of the interaction rule.

Finally, we established a **variational principle**: under the axioms of locality, covariance, and second-order dynamics, the interaction geometry is uniquely governed by

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}.$$

This equation provides a rigorous bridge between differential geometry and behavioral science, with $T_{\mu\nu}$ encoding reinforcement effects and perceptual forcing.

Future work will focus on:

- extending the manifold to higher-dimensional interaction spaces,
- specifying explicit forms of the interaction source tensor $T_{\mu\nu}$ for cognitive and social systems,
- exploring empirical validation within behavioral and neuroscientific datasets,
- investigating connections to matrix Ricci flow and optimal transport.

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